

## LIDGERWOOD, ND, USA

WMO: 727534

Lat: 46.100N

Lon: 97.150W

Elev: 351

StdP: 97.18

Time Zone: -6.00 (NAC)

Period: 86-96

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -27.4      | -24.9 | -32.9                         | 0.2 | -25.2 | -30.7 | 0.2 | -22.6 | 11.0                  | -0.7 | 9.8 | -1.9 | 1.6                   | 280  | 0.569 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.1                   | 33.3            | 21.9 | 31.4 | 20.5 | 29.6 | 19.5 | 23.5                | 31.2 | 22.2 | 29.2 | 20.9 | 27.7 | 4.2                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.9                            | 16.2 | 28.3 | 19.6 | 15.0 | 26.8 | 18.4 | 13.8 | 25.4 | 71.8          | 31.2 | 66.7 | 29.1 | 62.2 | 27.6 | 27.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 9.1               | 8.1  | 7.1 | -30.8                      | 36.9 | 2.8                | 2.8 | -32.8                                              | 38.9 | -34.5      | 40.6 | -36.1      | 42.1 | -38.2      | 44.2 |    |    |
|                   |      |     | -29.3                      | 24.9 | 2.0                | 1.7 | -30.7                                              | 26.1 | -31.9      | 27.1 | -33.0      | 28.0 | -34.4      | 29.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 6.1    | -11.0 | -9.0  | -2.0  | 6.6   | 15.0  | 19.8  | 21.1  | 20.2  | 14.8  | 7.3   | -2.3  | -8.5 |
|                                                                                  | DBStd       | 13.09  | 7.73  | 8.10  | 6.82  | 5.61  | 5.39  | 4.26  | 3.83  | 3.93  | 5.00  | 5.39  | 5.70  | 7.10 |
|                                                                                  | HDD10.0     | 2782   | 652   | 532   | 372   | 136   | 19    | 0     | 0     | 0     | 13    | 117   | 369   | 573  |
|                                                                                  | HDD18.3     | 4782   | 910   | 765   | 630   | 357   | 128   | 33    | 13    | 25    | 128   | 344   | 618   | 832  |
|                                                                                  | CDD10.0     | 1356   | 0     | 0     | 0     | 32    | 175   | 294   | 346   | 317   | 158   | 33    | 1     | 0    |
|                                                                                  | CDD18.3     | 314    | 0     | 0     | 0     | 3     | 26    | 77    | 100   | 84    | 23    | 2     | 0     | 0    |
|                                                                                  | CDH23.3     | 3563   | 0     | 0     | 0     | 65    | 338   | 877   | 1119  | 863   | 261   | 40    | 0     | 0    |
|                                                                                  | CDH26.7     | 1278   | 0     | 0     | 0     | 19    | 93    | 336   | 448   | 307   | 66    | 10    | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.0    | 3.1   | 3.2   | 3.3   | 3.5   | 3.6   | 3.0   | 2.5   | 2.4   | 2.6   | 2.8   | 3.1   | 2.9  |
| Precipitation                                                                    | PrecAvg     | 637    | 20    | 15    | 29    | 52    | 80    | 117   | 86    | 62    | 74    | 52    | 22    | 22   |
|                                                                                  | PrecMax     | 874    | 63    | 31    | 77    | 102   | 176   | 283   | 186   | 143   | 167   | 152   | 72    | 54   |
|                                                                                  | PrecMin     | 442    | 1     | 2     | 1     | 13    | 26    | 34    | 39    | 17    | 27    | 1     | 1     | 6    |
|                                                                                  | PrecStd     | 100    | 16    | 8     | 18    | 26    | 36    | 62    | 44    | 30    | 39    | 40    | 23    | 14   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 7.2   | 10.7  | 15.0  | 29.3  | 31.9  | 35.0  | 38.1  | 35.1  | 31.4  | 27.6  | 15.1  | 8.1  |
|                                                                                  |             | MCWB   | 3.0   | 5.0   | 8.4   | 15.9  | 18.4  | 21.6  | 23.3  | 23.7  | 18.6  | 15.7  | 9.4   | 4.1  |
|                                                                                  | 2%          | DB     | 3.9   | 7.4   | 11.1  | 24.9  | 29.4  | 32.5  | 33.8  | 32.1  | 28.5  | 23.0  | 10.9  | 4.8  |
|                                                                                  |             | MCWB   | 0.6   | 3.2   | 6.0   | 13.4  | 18.4  | 21.6  | 22.4  | 21.2  | 18.3  | 13.2  | 6.5   | 1.4  |
|                                                                                  | 5%          | DB     | 2.1   | 4.2   | 8.7   | 20.7  | 26.9  | 30.5  | 31.0  | 29.9  | 26.3  | 19.9  | 8.9   | 2.7  |
|                                                                                  |             | MCWB   | -0.5  | 1.3   | 5.0   | 11.9  | 16.5  | 20.4  | 21.0  | 19.6  | 16.9  | 11.7  | 4.9   | 0.1  |
|                                                                                  | 10%         | DB     | 0.2   | 2.1   | 5.9   | 16.4  | 25.0  | 28.3  | 29.0  | 27.9  | 24.1  | 16.4  | 6.3   | 0.9  |
|                                                                                  |             | MCWB   | -1.8  | -0.3  | 3.1   | 8.9   | 15.1  | 19.0  | 19.5  | 19.4  | 16.5  | 10.1  | 3.0   | -0.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.0   | 5.8   | 9.5   | 16.8  | 20.8  | 24.9  | 25.0  | 24.5  | 21.8  | 16.5  | 11.0  | 4.2  |
|                                                                                  |             | MCDB   | 6.6   | 10.2  | 14.2  | 29.2  | 29.9  | 32.3  | 33.3  | 31.4  | 28.5  | 24.5  | 14.4  | 8.5  |
|                                                                                  | 2%          | WB     | 0.8   | 3.3   | 6.8   | 14.2  | 19.1  | 22.7  | 23.7  | 22.8  | 20.1  | 14.6  | 7.2   | 1.6  |
|                                                                                  |             | MCDB   | 3.6   | 7.2   | 10.6  | 23.7  | 27.3  | 30.4  | 31.8  | 30.2  | 25.8  | 21.6  | 10.8  | 4.5  |
|                                                                                  | 5%          | WB     | -0.3  | 1.4   | 5.0   | 11.9  | 17.7  | 21.3  | 22.4  | 21.6  | 18.6  | 12.6  | 4.9   | 0.3  |
|                                                                                  |             | MCDB   | 2.1   | 4.2   | 8.4   | 19.9  | 25.4  | 28.7  | 29.5  | 27.8  | 24.4  | 19.1  | 8.6   | 2.6  |
|                                                                                  | 10%         | WB     | -1.7  | 0.0   | 3.2   | 9.7   | 16.3  | 19.9  | 20.8  | 20.3  | 17.1  | 10.5  | 3.0   | -0.8 |
|                                                                                  |             | MCDB   | 0.2   | 2.3   | 5.8   | 15.9  | 23.2  | 27.0  | 28.0  | 26.3  | 22.8  | 16.0  | 6.3   | 1.0  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.0  | 9.8   | 9.3   | 12.4  | 12.8  | 12.3  | 12.1  | 12.7  | 13.4  | 13.0  | 9.1   | 9.2  |
|                                                                                  |             | MCDBR  | 11.1  | 12.8  | 12.7  | 18.6  | 15.7  | 14.7  | 14.4  | 14.7  | 16.3  | 18.6  | 12.7  | 11.5 |
|                                                                                  |             | MCWBR  | 9.0   | 10.0  | 9.2   | 10.4  | 7.9   | 7.2   | 6.6   | 7.4   | 8.7   | 11.0  | 9.4   | 9.1  |
|                                                                                  | 5% WB       | MCDBR  | 11.2  | 12.6  | 12.0  | 17.5  | 13.2  | 12.9  | 12.4  | 12.2  | 13.2  | 16.6  | 12.2  | 11.0 |
|                                                                                  |             | MCWBR  | 9.3   | 9.9   | 8.8   | 10.2  | 7.3   | 6.9   | 6.2   | 7.2   | 7.5   | 9.9   | 9.2   | 8.8  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.273  | 0.281 | 0.303 | 0.355 | 0.377 | 0.390 | 0.385 | 0.395 | 0.357 | 0.320 | 0.284 | 0.263 |      |
|                                                                                  | taud        | 2.363  | 2.380 | 2.444 | 2.338 | 2.351 | 2.360 | 2.367 | 2.331 | 2.420 | 2.497 | 2.501 | 2.422 |      |
|                                                                                  | Ebn at Noon | 835    | 903   | 931   | 902   | 890   | 876   | 876   | 852   | 858   | 847   | 823   | 812   |      |
|                                                                                  | Edh at Noon | 73     | 88    | 96    | 117   | 120   | 120   | 117   | 117   | 98    | 78    | 64    | 62    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.71   | 2.83  | 3.84  | 4.89  | 5.52  | 6.17  | 6.50  | 5.45  | 4.14  | 2.60  | 1.66  | 1.24  |      |
|                                                                                  | RadStd      | 0.18   | 0.30  | 0.36  | 0.38  | 0.38  | 0.38  | 0.26  | 0.48  | 0.27  | 0.35  | 0.18  | 0.13  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)  | (e)                     | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (28 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page